
MULTILINGUAL GSM SYMBOLIC

Anonymous authors

Paper under double-blind review

ABSTRACT

Why and to what degree does capabilities transfer across languages? Answering this allow for better utilization of resources in both high-resource and low-resource communities as it allows for more efficient evaluation and guides model development. Utilizing mathematics as a underlying computation remains constant, we construct a dataset of item-level matches word problems across a diverse set of languages. To prevent overfitting, ensure generalization and ensure high quality samples we utilize symbolic templates, that allow millions of variants to be generated from a single example.

Utilizing this dataset we evaluate a range of models families covering size, multilinguality, and reasoning capabilities and fit a generalized mixed-effect models to find that language performance is determined most notably by model size (**effect, p-value**), language prevalence in the training data (**effect, p-value**) and to a lesser extent influenced by tokenizer fertility (**effect, p-value**), the number of syntax features in common with English (**effect, p-value**).

To allow for reproducibility we share the code¹ and dataset publicly² and contribute the created dataset to existing evaluation frameworks³.

1 INTRODUCTION

Why and to what degree does capabilities transfer across languages? There is a need for a better understanding of transfer both for evaluation, training and architectural developments. If we understand what factors predict transfer, we can avoid costly exhaustive evaluation across all language pairs and model developers can directly target the factors that limit performance in underrepresented languages. To explore what determines transfer, it is important to isolate central-varying components that can contribute to the variance across languages, for instance, while evaluating models on English and Chinese mathematics exam, tell you something about a models relative improvements it doesn't tell you anything about the effect of language as different exams varies on multiple axis formatting and difficulty, making it hard to isolate the effect of language itself. Similarly, for domains like medical question answering the answers might differ between languages as medical practices vary. For instance, recommended drug choices and pre-prescription tests can differ by region because of population-level differences in adverse-reaction risk. Thus, when studying transfer, mathematics provides a useful testbed as the underlying computation remains constant across languages, leaving the language as the remaining unit of variance.

While some datasets already exists (Shi et al., 2022; Singh et al., 2025) which covers mathematics and has item-level matches samples across languages, these were both popular and static making it hard to differentiate isolate capabilities from leakage. To combat this we utilize the template approach introduces in GSM-Symbolic (Mirzadeh et al., 2024), where a singular template

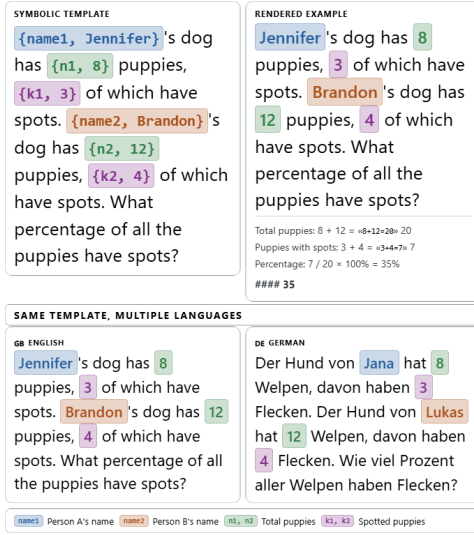
In this paper we introduce Multilingual GSM-Symbolic, a benchmark of parallel math question templates across N languages, translated with extensive manual review and automatic validation.

We use this benchmark to study two classes of predictors across a comprehensive set of models. The first is model level predictors such as size, pretraining language distribution, tokenizer architecture and family. We also study language level factors such as syntactic features. Prior work has shown that

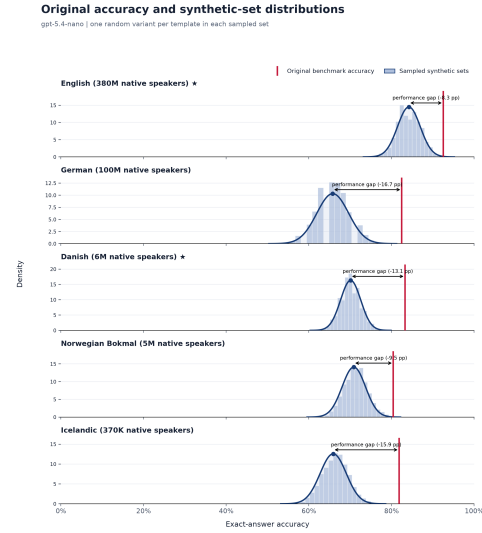
¹Anonymized

²Anonymized

³Anonymized



(a) Multilingual GSM Symbolic Templates: A single problem exists in multiple languages that allow generation of novel localized samples.



(b) GPT 5.4-nano performance on original vs synthetic.

Figure 1: Comparison of template generation and cross-language performance.

these factors matter: size has been shown to reduce transfer gaps (Shi et al., 2022; Barua et al., 2026), and pretraining token distribution and syntactic similarity impact performance (Shi et al., 2022; Barua et al., 2026). However, model and language factors have rarely been studied jointly across languages using modern reasoning models.

These contributions mark a shift from headline multilingual accuracy to an in depth analysis of why transfer succeeds or fails

if we need evaluation across all languages or whether some language task pairs can be estimates it determined what data we should collect for training, how we should design model components such as the tokenizer.

While previous work have examined transfer ... there has been a lack of work examining both model factors (like tokenizers) and language factors (such as syntactic features) at the same time. Existing work has also focused on a limited selection of families and sizes and has primarily been conducted on older LLM architectures.

In this work we seek to systematically examine transfer across languages and to do so construct a highly controlled dataset of symbolic mathematics template across N languages.

This dataset enables us to...

understand what factors predict transfer

2 RELATED WORK

2.1 MULTILINGUAL MATHEMATICAL REASONING

Mathematics is useful for studying reasoning as it requires models to utilize multi step reasoning and its answers can be objectively verified (Wen et al., 2026; Cobbe et al., 2021). Additionally, mathematics fundamentally remains consistent across cultural contexts, compared to e.g. social or medical reasoning that are influenced by the context in which it operated (McNamara et al., 2019; Napier et al., 2014). Grade school math is especially useful because the underlying mathematical operations are elementary, placing greater emphasis on whether models can constructing the underlying reasoning procedure than their computational abilities. GSM8K (Cobbe et al., 2021) and its variants (Shi et al., 2022; Vendrow et al., 2025) comprise a family of benchmark consisting of grade school

level arithmetic word problems requiring multi-step reasoning. GSM8K is limited by its reliance on a static data set: each problem appears with fixed text and values. This makes it difficult to separate underlying reasoning performance from sensitivity to surface form (Mirzadeh et al., 2024) and memorization enabled by potential contamination (Zhang et al., 2024). GSM-Symbolic (Mirzadeh et al., 2024) turns GSM8K questions into templates, and demonstrates that surface-level substitutions of names and numbers significantly reduce performance, showing the limited nature of static datasets. Efforts have also been made to extend math benchmarks to other languages. MGSM (Shi et al., 2022) translates a 250-question subset of GSM8K into multiple languages. While MGSM is useful for comparing headline performance across languages, its size and static nature are limitations.

Variations of MGSM exist to address these problems.

MGSM-Pro (Xu et al., 2026) extends MGSM using templates, introducing symbolic and irrelevant context sets, but it inherits template issues from GSM8K and does not release the dataset used in the paper or template generation code. We include extensive automated and human review to ensure question quality and open source all code.⁴

mGSM-Symbolic (Ranaldi and Pucci, 2025) statically translates GSM-Symbolic question instances across languages, but does not use templates meaning the dataset is static. There is limited description of any validation of translations or templates.⁵

Other benchmarks measure broader knowledge and reasoning ability. Global MMLU (Singh et al., 2025) is a translation of the popular multidisciplinary English-only benchmark MMLU. It is limited by the fact that LLMs frequently use choice-based shortcuts on multiple-choice questions which means that these datasets measure discrimination ability rather than reasoning (Chandak et al., 2025). Global MMLU also requires culturally specific knowledge on many questions which can be confounding. By contrast, grade school math reduces cultural confounds, providing a task that is more consistent between languages than broad-knowledge benchmarks (Hendrycks et al., 2021).

Here is another possibly related paper doing multilingual reasoning benchmarks. The plan is to work with the creators and run our suite on their benchmark and compare rankings. It supports 7 languages. <https://arxiv.org/pdf/2511.03553>

MGSM-Pro (Xu et al., 2026) extends the GSM-Symbolic templates and does something insightful, but has these limitations. Here we do ABC to explore DEF. **Isaac: Example structure.**

2.2 WHAT DETERMINES TRANSFER?

MGSM investigated the impact of model scale on transfer, finding that relative transfer efficiency increases with model size within the same family. Work on pretraining for reasoning models has also demonstrated that increasing model size reduces relative transfer gaps. (Barua et al., 2026) These existing transfer studies only explore single model families which is a concern as post training studies have shown model selection to be a significant factor in transfer efficiency. (Anonymous, 2026)

Other work has examined the impact of language features on transfer. Experiments on transformer classifiers have shown significant correlation between target-language performance and pretraining token frequency as well as language similarity to English. (Blevins and Zettlemoyer, 2022; Lauscher et al., 2020)

(Chang et al., 2023) demonstrated that in small LLMs (max 45M parameters) adding multilingual data improved non-English performance and that this improvement was based on the syntactic similarity of the added language. This suggests that syntactic similarity is important, but these results may not generalize to modern LLMs in the tens of billions of parameters.

(Barua et al., 2026) showed that in post training of reasoning models, the addition of target-language data had a high impact on performance, suggesting that language distribution in training corpora is important.

⁴MGSM-Pro first appeared on arXiv on January 29, 2026 after our initial public release of the repository and dataset on February 26, 2025.

⁵MGSM Symbolic created their repository on February 21, 2025 but did not publish any data, code or documentation until May 1, 2025, after our initial public release of the repository and dataset on February 26, 2025

Dataset	Languages	Symbolic	Code	Extensible	Validation			
					Error Analysis	Translation	Localization	Validity
multilingual-gsm-symbolic (ours)	15 (33)	✓	✓	✓	✓	✓	✓	✓
Previous work								
MGSM (Shi et al., 2022)	11					✓	✓	
GSM-Symbolic (Mirzadeh et al., 2024)	1	✓						
GSM8K-Platinum (Vendrow et al., 2025)	1				✓			
Parallel work								
mGSM-Symbolic (Ranaldi and Pucci, 2025)	11					✓		
MGSM-Pro (Xu et al., 2026)	9	✓				✓	✓	✓

Table 1: Comparison of GSM-style mathematical reasoning benchmarks. The parenthesis refer to languages that have not been human validated yet. For more see4 Our dataset combines multilingual templates, controllable variation and extensibility with extensive validation. Error analysis refer to the process of running a model and correcting ambiguous or incorrect questions. Extensible as defined by (Enevoldsen et al., 2025), indicate that there is a process for adding language, extending or correcting the dataset. Validity denote automated tests that ensures the quality of the generated examples.

This prior work suggests that both model and language factors impact transfer. However, these factors have rarely been jointly studied in modern LLMs.

Our Dataset

Our dataset addresses previous limitations in multilingual benchmarks in answer choice, cultural knowledge, size and contamination by creating parallel symbolic variants across languages. This design allows us to hold the underlying reasoning constant while varying surface form, numerical values and language. As a result, we can move beyond simply comparing accuracy across languages and instead analyze in detail what factors predict the success of the transfer.

Other multilingual reasoning benchmarks

<https://arxiv.org/pdf/2511.03553> <https://bsky.app/profile/adam-gaber.bsky.social/post/3mqjduaaxgk2x>

3 METHODS

4 DATASET CONSTRUCTION

Source Data and Template Process We construct Multilingual GSM-Symbolic based on GSM-Symbolic. As the authors of GSM-Symbolic provide templates but not a parser, we implement and release our own parser, extending the syntax of GSM-Symbolic templates by adding additional functions to improve multilingual support.

Target Language Selection

We select languages from a diverse set of language families and scripts, with languages of varying resource level within each family/script. The diversity of resource level within families enables comparison between languages that share many linguistic properties but differ in their prevalence in pretraining datasets.

Translation Process

To begin the translation process LLMs were used to translate templates. These templates were then all manually reviewed by expert native-speaker annotators. Each template includes a creation string documenting the source language of the translation and the model used

We localize currencies, units, cultural references and numerals in templates. Localization is used as literal translation can introduce confounds. A problem containing foreign currency, units, or cultural concepts may be harder in a language because such problems are non-naturalistic and out-of-distribution. We therefore define item level equivalence at the level of the underlying reasoning structure and mathematical operations, which remain constant, rather than at the level of exact problem text which introduces confounding translation artifacts.

Validation

Because prior work has identified ambiguities and errors in GSM8K (Vendrow et al., 2025) we apply a series of verifications to templates on top of those done by the authors of GSM-Symbolic. These checks are used on English and translated templates, preventing English template errors from propagating across languages while also catching errors introduced during translation.

After the initial translation and review, we generate 20 instances of each problem and manually review all templates that Claude Opus 4.8 xhigh answers incorrectly, reviewing reasoning traces to understand failure points. This process is useful for catching unclear wording. For example, one question from GSM8K asks: "How much more likely is it that he rolls a number greater than 3 than that he rolls two even numbers in a row?". Because both a relative interpretation (200 percent) and an absolute interpretation (25 percentage points) are valid, we clarify the question by adding "expressed as percentage points."

All templates in all languages are also subject to automated checks. First they are rendered using the original values to ensure that they are a match with the original question and that values are in the domain and satisfy all conditions, addressing mismatch issues in GSM-Symbolic (Ivanova et al., 2025). We then evaluate each symbolic equation, $\{\llbracket \text{lhs}=\text{rhs} \rrbracket\}$, for equality. These checks are designed to catch both mathematical inconsistencies and rendering errors that could otherwise make original and synthetic templates incomparable.

Additional Languages

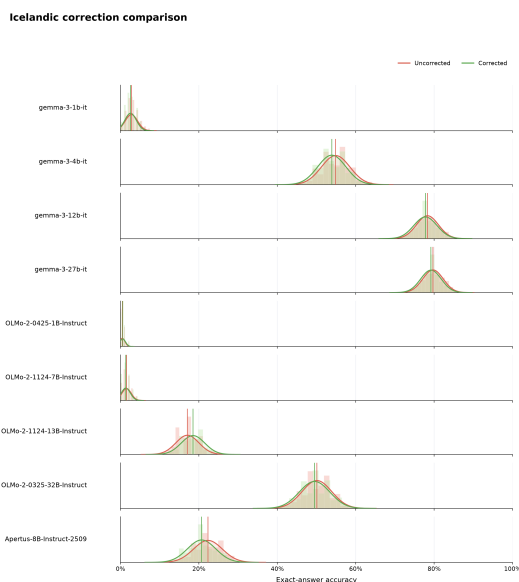


Figure 2: Accuracy on machine-generated vs human-validated icelandic templates

In Icelandic templates, accuracy on translated templates seems to approximate accuracy on verified templates, with models scoring on average 30 percent on unverified and 31 percent on verified. This suggests that translated templates can be used to extend this benchmark to additional languages. To extend the usability of our data for future we translate an additional 18 languages to expand our benchmark to include all of the top 30 languages as well as all of the languages in EuroEval. This allows us to contribute MGSM symbolic to EuroEval, a standard evaluation suite. We do not use these translations in the paper, to reduce potential bias from using machine translation. We include these in a package to allow future contributors to validate these and version the package to enable reproducibility and traceability.

5 MODELLING

The statistical modelling of the results. (see slack channel canvas)

We choose to use common crawl page count as a resource level proxy as common crawl is widely used in pretraining LLMs and it has a lot of the internet so it likely represents overall average frequencies. Other methods like token or word could be marginally more accurate but betokening the entire corpus is computationally infeasible and word counts in Chinese are different than those in english because Chinese uses no spaces.

5.1 MODEL SELECTION

Our goal is to select a suite of models that makes the determinants of transfer estimable. We evaluate instruction-tuned, open-weight models organized in a family $\times$ size grid, allowing us to estimate the effect of scale within a model family while controlling for family-level differences such as architecture, tokenizer, and pretraining mixture. We prioritized model families that included a wide range of model sizes to enable performance comparison across within-family size changes.

We restrict the suite to instruction-tuned variants, since reliable performance on GSM-style problems benefits from instruction-following and because these instruction-tuned variants better reflect the models used in real-world applications (Chung et al., 2022). Our estimates therefore characterize transfer in the *aligned* model and do not separately attribute it to the pretraining versus alignment stages. We also include a selection of reasoning oriented models.

We further restrict to open-weight models so that each model’s tokenizer is available for computing fertility-based language features and so its generated reasoning traces can be inspected for analysis.

Although Mixture of Experts models are becoming increasingly prevalent (Cai et al., 2025), we do not include any Mixture of Experts models to ensure size comparisons do not compare between Mixture of Experts and dense models. We leave it to future work to study the relationship between active and total parameters and cross lingual performance in Mixture of Experts models.

Three of our model families, EuroLLM, Apertus and OLMo have fully open pretraining datasets with varying language distributions. OLMo is over 90% English while Apertus and EuroLLM are 50% and 40% respectively. Access to the exact pretraining token count lets us understand the correlation between pretraining token counts and language performance.

6 RESULTS

Some initial results shared in slack channel

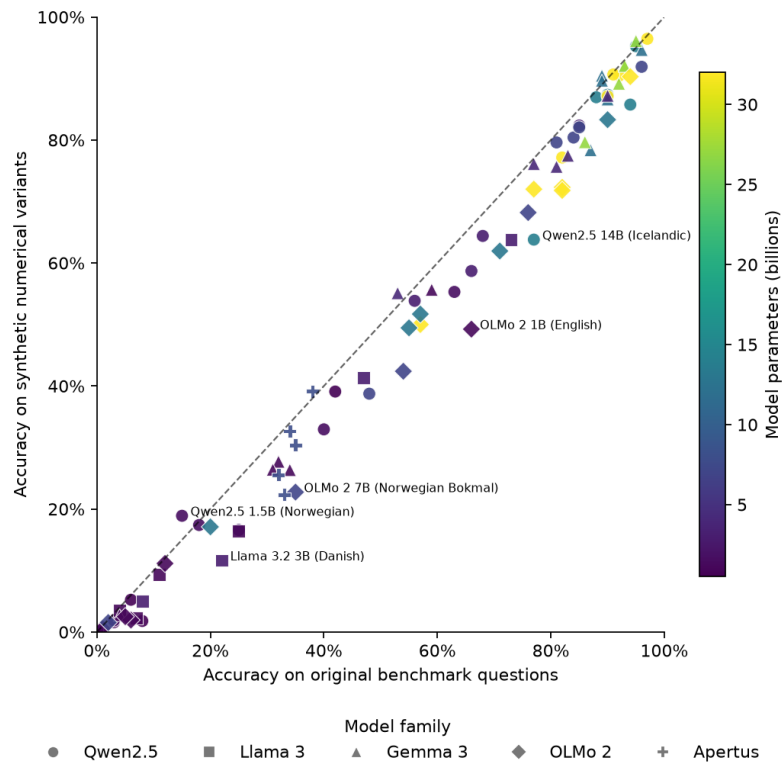


Figure 3: Original Question Accuracy vs Synthetic Variant Accuracy

7 DISCUSSION

CONCLUSION

LIMITATIONS

Size of the dataset. Reviewers for MSGM criticized them for only having 250 examples but they showed a high correlation between performance of that subset and the whole set which could be a good approach for us too. We could also see correlation between 10, 25, 50, 75 etc and see how to correlation increases w/ size to show we're in a good spot when it comes to diminishing returns.MGSM

In the original GSM-Symbolic paper, there were some concerns that due to the use of number ranges in symbolic templates that the synthetic questions were more difficult. The authors of GSM-Symbolic did extensive experiments showing limited differences in performance between questions from popular math benchmarks where the final answer had 1,2,3, or 4 digits.

The problems in the dataset are relatively simple, requiring only grade-school math, so this benchmark may reach the point of saturation in large frontier models. To extend the lifespan of this dataset adversarial sampling approaches like the beam search proposed in (Mondorf et al., 2026) could be used to select the most difficult problems in a computationally efficient manner to create challenging subsets that even frontier models struggle with.

We explore transfer across languages - but could also examine transfer across domains.

<https://openreview.net/forum?id=AjXkrZIVjB¬eId=HccaIYHvO5>

Interesting blog post about the limitations of GSM-Symbolic post

REFERENCES

- Anonymous. 2026. How well does mathematical reasoning transfer across languages in english-centric LLMs? In *Submitted to ACL Rolling Review - March 2026*. Under review.
- Josh Barua, Seun Eisape, Kayo Yin, and Alane Suhr. 2026. Long chain-of-thought reasoning across languages. *Preprint*, arXiv:2508.14828.
- Allan Bernardo. 2002. Language and mathematical problem solving among bilinguals. *The Journal of psychology*, 136:283–97.
- Terra Blevins and Luke Zettlemoyer. 2022. Language contamination helps explain the cross-lingual capabilities of english pretrained models. *Preprint*, arXiv:2204.08110.
- Weilin Cai, Juyong Jiang, Fan Wang, Jing Tang, Sunghun Kim, and Jiayi Huang. 2025. A survey on mixture of experts in large language models. *IEEE Transactions on Knowledge and Data Engineering*, page 1–20.
- Nikhil Chandak, Shashwat Goel, Ameya Prabhu, Moritz Hardt, and Jonas Geiping. 2025. Answer matching outperforms multiple choice for language model evaluation. *Preprint*, arXiv:2507.02856.
- Tyler A. Chang, Catherine Arnett, Zhuowen Tu, and Benjamin K. Bergen. 2023. When is multilinguality a curse? language modeling for 250 high- and low-resource languages. *Preprint*, arXiv:2311.09205.
- Hyung Won Chung, Le Hou, Shayne Longpre, Barret Zoph, Yi Tay, William Fedus, Yunxuan Li, Xuezhi Wang, Mostafa Dehghani, Siddhartha Brahma, Albert Webson, Shixiang Shane Gu, Zhuyun Dai, Mirac Suzgun, Xinyun Chen, Aakanksha Chowdhery, Alex Castro-Ros, Marie Pellat, Kevin Robinson, and 16 others. 2022. Scaling instruction-finetuned language models. *Preprint*, arXiv:2210.11416.
- Karl Cobbe, Vineet Kosaraju, Mohammad Bavarian, Mark Chen, Heewoo Jun, Lukasz Kaiser, Matthias Plappert, Jerry Tworek, Jacob Hilton, Reiichiro Nakano, Christopher Hesse, and John Schulman. 2021. Training verifiers to solve math word problems. *arXiv preprint arXiv:2110.14168*.
- Clément Dumas, Chris Wendler, Veniamin Veselovsky, Giovanni Monea, and Robert West. 2025. Separating tongue from thought: Activation patching reveals language-agnostic concept representations in transformers. *Preprint*, arXiv:2411.08745.
- Kenneth Enevoldsen, Kristian Nørgaard Jensen, Jan Kostkan, Balázs Szabó, Márton Kardos, Kirten Vad, Johan Heinsen, Andrea Blasi Núñez, Gianluca Barmina, Jacob Nielsen, and 1 others. 2025. Dynaword: From one-shot to continuously developed datasets. In *Proceedings of the Language Resources and Evaluation Conference (LREC 2026)*.
- Dan Hendrycks, Collin Burns, Steven Basart, Andy Zou, Mantas Mazeika, Dawn Song, and Jacob Steinhardt. 2021. Measuring massive multitask language understanding. *Preprint*, arXiv:2009.03300.
- Desi R Ivanova, Ilija Ilievski, and Momchil Konstantinov. 2025. Towards more rigorous evaluations of language models. In *ICLR Blogposts 2025*. <https://iclr-blogposts.github.io/2025/blog/towards-more-rigorous-llm-evals/>.
- Anne Lauscher, Vinit Ravishankar, Ivan Vulić, and Goran Glavaš. 2020. From zero to hero: On the limitations of zero-shot cross-lingual transfer with multilingual transformers. *Preprint*, arXiv:2005.00633.
- Rita Anne McNamara, Aiyana K. Willard, Ara Norenzayan, and Joseph Henrich. 2019. Weighing outcome vs. intent across societies: How cultural models of mind shape moral reasoning. *Cognition*, 182:95–108.
- Iman Mirzadeh, Keivan Alizadeh-Vahid, Hooman Shahrokhi, Oncel Tuzel, Samy Bengio, and Mehrdad Farajtabar. 2024. Gsm-symbolic: Understanding the limitations of mathematical reasoning in large language models. *ArXiv*, abs/2410.05229.

-
- Philipp Mondorf, Samuel J. Bell, Jesse Dodge, and Dieuwke Hupkes. 2026. Lpds: Evaluating llm robustness through logic-preserving difficulty scaling. *Preprint*, arXiv:2605.15393.
- A. David Napier, Clyde Ancarno, Beverley Butler, Joseph D. Calabrese, Angel Marie Chater, Helen J. Chatterjee, François Guesnet, Rob Horne, Stephen Jacyna, Sushrut Jadhav, Alison Macdonald, Ulrike Neuendorf, Aaron Parkhurst, Rodney J. Reynolds, Graham Scambler, Sonu Shamdasani, Sonia Smith, Jakob Stougaard-Nielsen, Linda J M Thomson, and 8 others. 2014. Culture and health. *Lancet*, 384 9954:1607–39.
- Leonardo Ranaldi and Giulia Pucci. 2025. Multilingual reasoning via self-training. In *Proceedings of the 2025 Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, pages 11566–11582, Albuquerque, New Mexico. Association for Computational Linguistics.
- Freda Shi, Mirac Suzgun, Markus Freitag, Xuezhi Wang, Suraj Srivats, Soroush Vosoughi, Hyung Won Chung, Yi Tay, Sebastian Ruder, Denny Zhou, Dipanjan Das, and Jason Wei. 2022. Language models are multilingual chain-of-thought reasoners. *ArXiv*, abs/2210.03057.
- Shivalika Singh, Angelika Romanou, Clémentine Fourrier, David I. Adelani, Jian Gang Ngui, Daniel Vila-Suero, Peerat Limkonchotiwat, Kelly Marchisio, Wei Qi Leong, Yosephine Susanto, Raymond Ng, Shayne Longpre, Wei-Yin Ko, Sebastian Ruder, Madeline Smith, Antoine Bosselut, Alice Oh, Andre F. T. Martins, Leshem Choshen, and 5 others. 2025. Global mmlu: Understanding and addressing cultural and linguistic biases in multilingual evaluation. *Preprint*, arXiv:2412.03304.
- Joshua Vendrow, Edward Vendrow, Sara Beery, and Aleksander Madry. 2025. Do large language model benchmarks test reliability? *Preprint*, arXiv:2502.03461.
- Xumeng Wen, Zihan Liu, Shun Zheng, Shengyu Ye, Zhirong Wu, Yang Wang, Zhijian Xu, Xiao Liang, Junjie Li, Ziming Miao, Jiang Bian, and Mao Yang. 2026. Reinforcement learning with verifiable rewards implicitly incentivizes correct reasoning in base LLMs. In *The Fourteenth International Conference on Learning Representations*.
- Chris Wendler, Veniamin Veselovsky, Giovanni Monea, and Robert West. 2024. Do llamas work in english? on the latent language of multilingual transformers. *Preprint*, arXiv:2402.10588.
- Tianyi Xu, Kosei Uemura, Alfred Malengo Kondoro, Tadesse Destaw Belay, Catherine Nana Nyaah Essuman, Ifeoma Okoh, Ganiyat Afolabi, Ayodele Awokoya, and David Ifeoluwa Adelani. 2026. Mgsn-pro: A simple strategy for robust multilingual mathematical reasoning evaluation. *Preprint*, arXiv:2601.21225.
- Hugh Zhang, Jeff Da, Dean Lee, Vaughn Robinson, Catherine Wu, Will Song, Tiffany Zhao, Pranav Raja, Charlotte Zhuang, Dylan Slack, Qin Lyu, Sean Hendryx, Russell Kaplan, Michele Lunati, and Summer Yue. 2024. A careful examination of large language model performance on grade school arithmetic. *Preprint*, arXiv:2405.00332.

A EXAMPLE APPENDIX

Reasoning vs Non Reasoning: Model selection to reduce transfer gap within a given compute budget

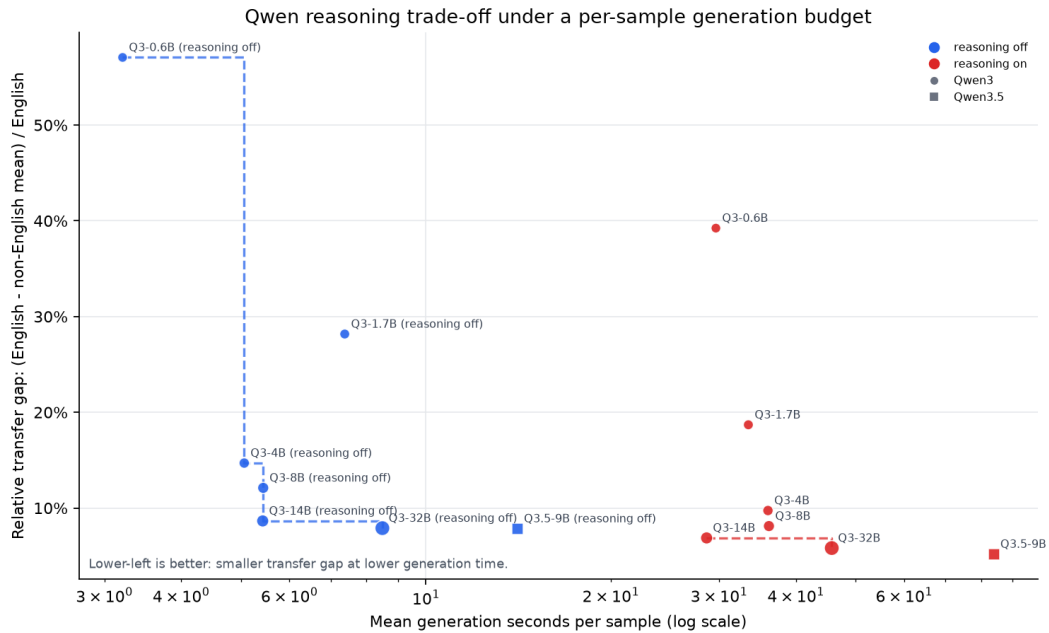


Figure 5: Enter Caption

When selecting models to reduce the transfer gap between English and other languages, one important choice is to use reasoning or non-reasoning models. We find that with a lower compute budget (measured in seconds to generate each sample on an Nvidia Blackwell B200) that non-reasoning models are effective in reducing gaps while reasoning models can provide the greatest absolute reduction. Enabling reasoning on average reduce transfer gaps by 24 percent while increasing generation time per question by 370 percent.

It would be interesting to explore the difference between very large models without reasoning and smaller models with reasoning when working with a higher compute budget, but there is a lack of model families with toggelable reasoning that span the necessary breadth of sizes to perform this analysis.

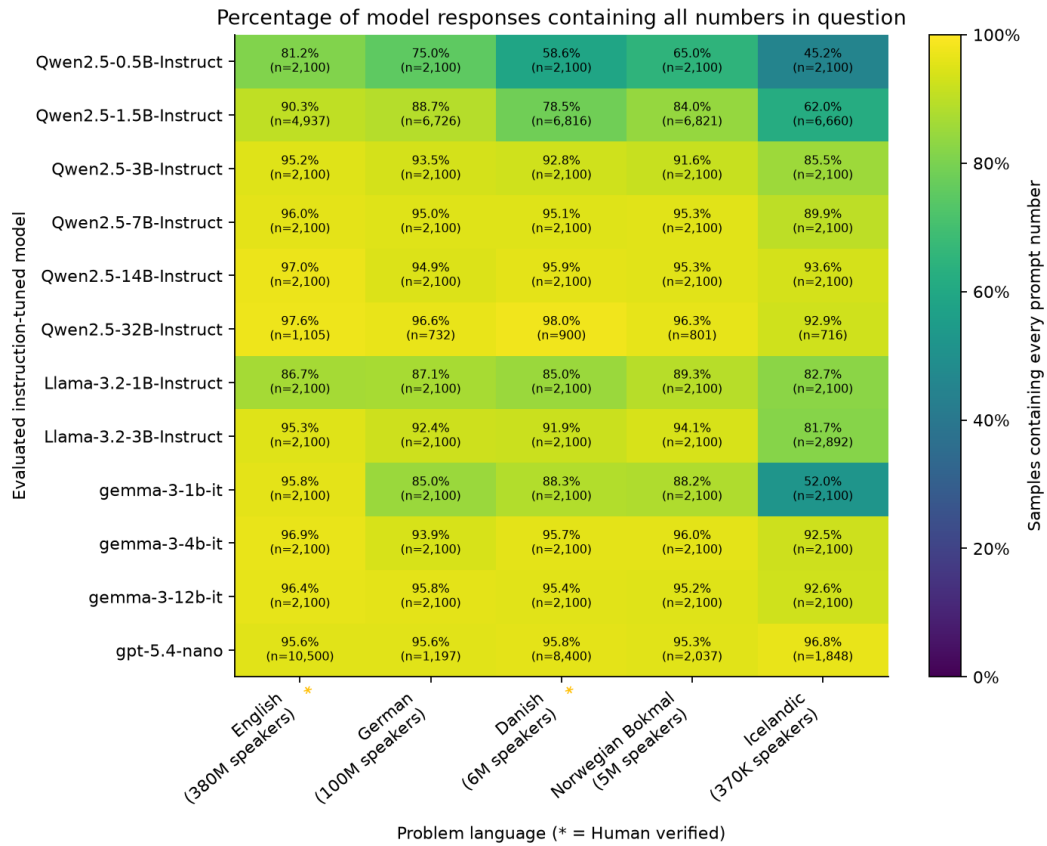


Figure 6: Percentage of model responses containing all numbers in question

Understanding whether incorrect responses arise from failures to correctly extract information in the prompt or from errors correctly applying reasoning procedure is important for understanding transfer gaps. If LLMs fail primarily due to language extraction then language understanding may be the limiting factor while if LLMs correctly extract numbers but fail while doing math this reflects instability in reasoning across languages.

We therefore define number coverage: the percent of numbers from the prompt appearing in the reasoning trace or final response. This metric is motivated by human studies of multilingual math problem solving which distinguish understanding the problem representation from executing the mathematical procedure (Bernardo, 2002).

In our dataset, there are very few problems that contain any irrelevant numbers so using all the numbers in the prompt is usually necessary to get the correct answer. We equate numeric and written forms of numbers ("three" and 3) when doing this comparison. Number coverage is an imperfect metric as models may copy numbers without understanding them and prompt numbers may appear in intermediate results. We therefore treat number coverage as an upper bound rather than a direct measure of semantic understanding.

We find an r squared of 0.57 between number coverage and accuracy suggesting that while information extraction can explain a substantial portion of performance variation it does not explain all multilingual errors. This motivates additional analysis into how models fail after extracting relevant quantities, for example by applying incorrect operations, misunderstanding relationships between quantities or making arithmetic errors.

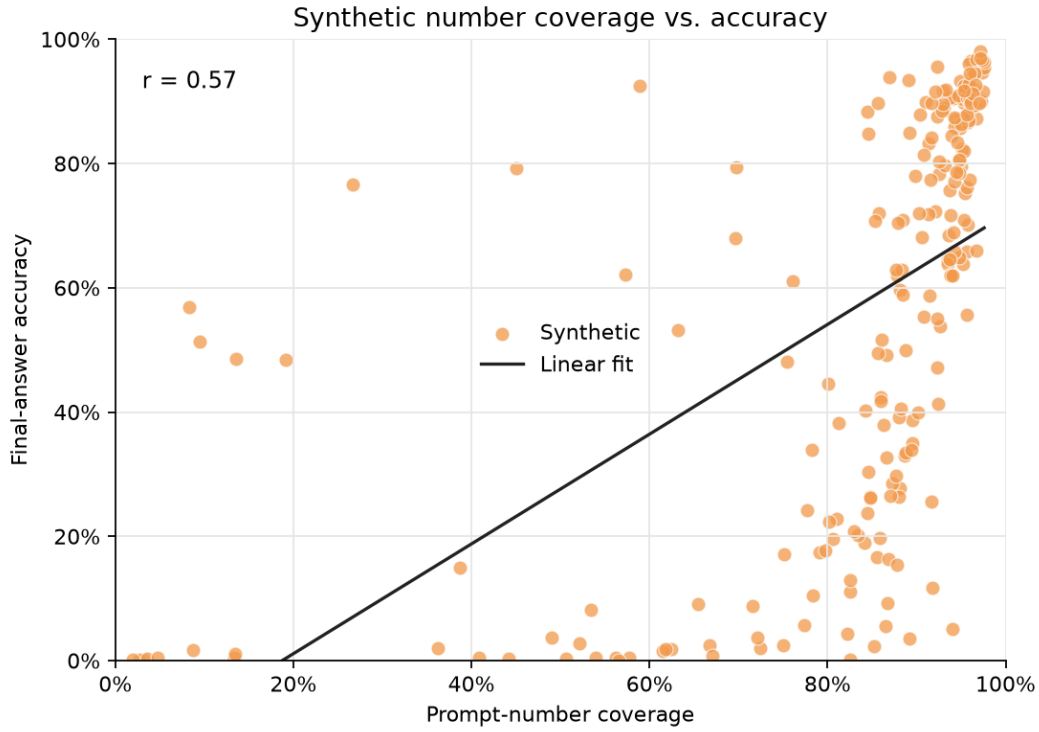


Figure 7: Enter Caption

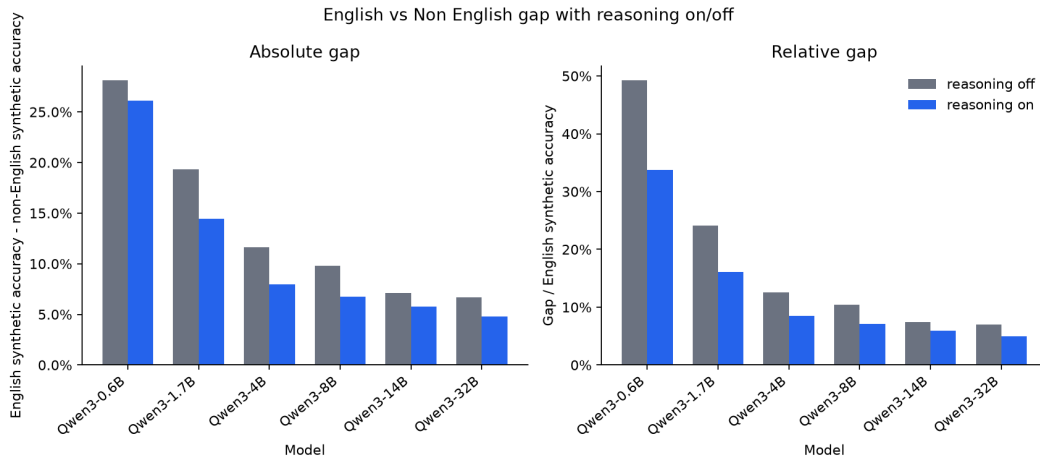


Figure 8: Transfer gap with reasoning enabled vs disabled

Reasoning models are increasingly common. We show that enabling reasoning in models reduces transfer gaps. We do not specify a language for models to reason in so this might be because models frequently choose to internally reason in English, leading to more uniform performance across languages, compared to reasoning off scenarios where they respond entirely in the target language (Shi et al., 2022). It has been observed that models map multilingual inputs into internal representations before producing an answer (Dumas et al., 2025; Wendler et al., 2024) and it could be the case that this is more pronounced in reasoning models, reducing transfer gaps.

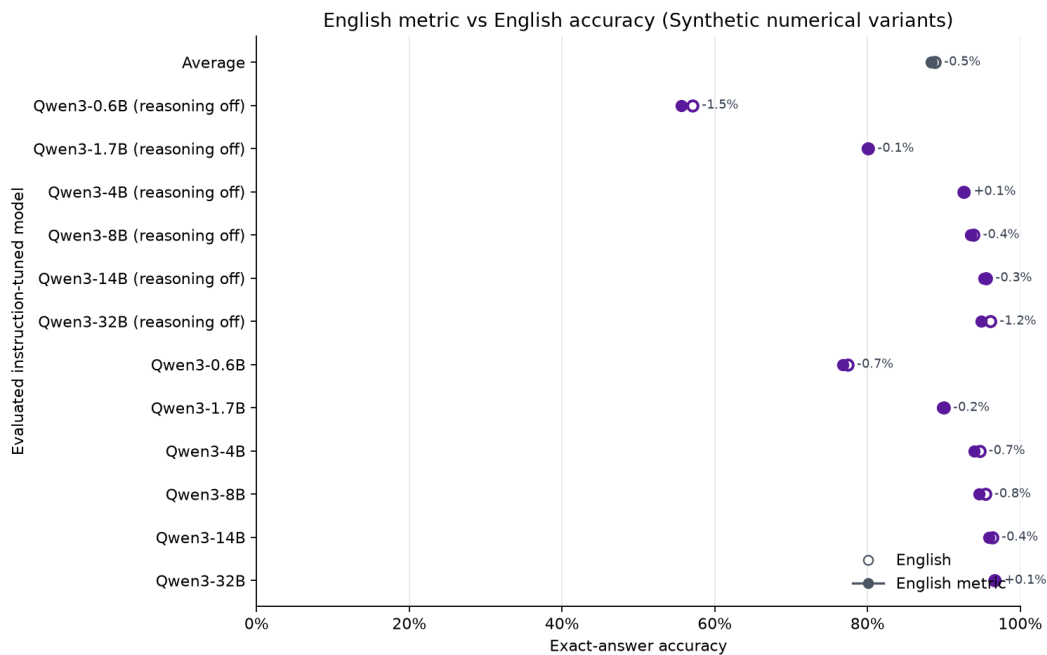


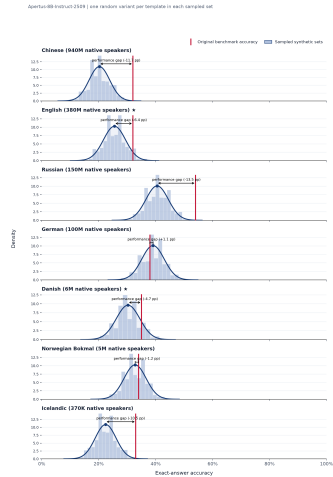
Figure 9: Performance on English templates using metric vs imperial units

To understand if multilingual performance gaps are due to non-English templates using metric units as opposed to the imperial units in English templates, we construct a metric version of our English dataset. Comparing performance on the metric and non metric versions of English we see an average difference of 0.5 percent, but this is not significant within the 95 percent CI. We report the performance difference on the subset of templates that use metric/non metric units but this subset only represents 25 percent of templates so the slight performance differences are diluted when considering the whole dataset. This suggests that the change in units cannot explain a significant portion of the performance difference between english and non-English templates.

810
811
812
813
814
815
816
817
818
819
820
821
822
823
824
825
826
827
828
829
830
831
832
833
834
835
836
837
838
839
840
841
842
843
844
845
846
847
848
849
850
851
852
853
854
855
856
857
858
859
860
861
862
863

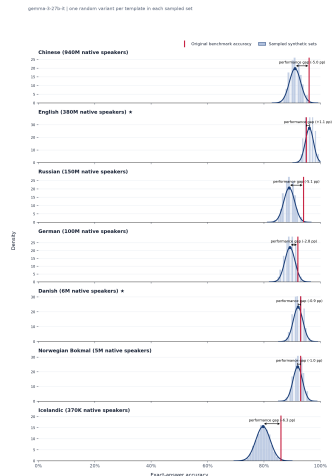
B RIDGELINE PLOTS

Original accuracy and synthetic-set distributions



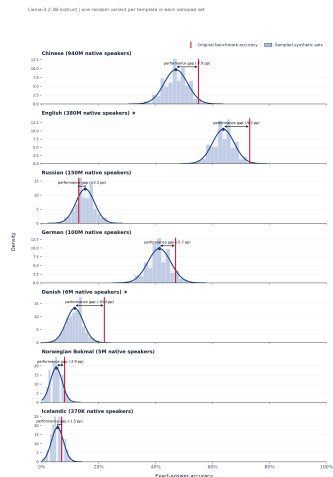
apertus-8b-instruct-2509.png

Original accuracy and synthetic-set distributions



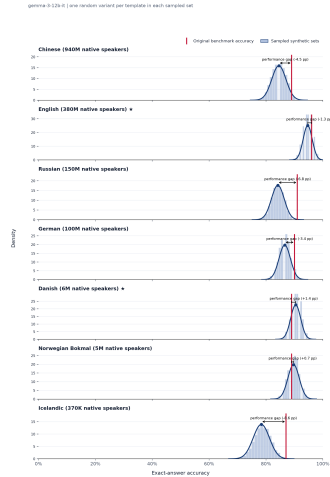
gemma-3-27b-it.png

Original accuracy and synthetic-set distributions



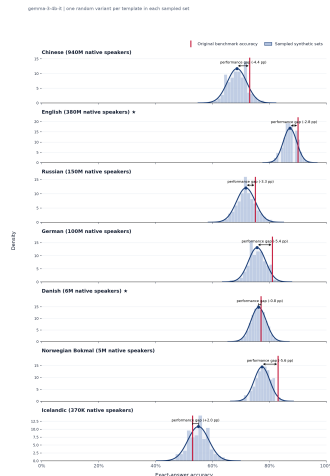
llama-3.2-3b-instruct.png

Original accuracy and synthetic-set distributions



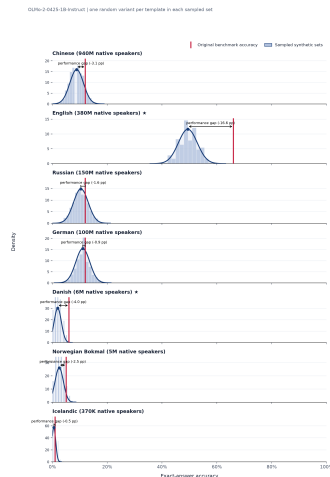
gemma-3-12b-it.png

Original accuracy and synthetic-set distributions



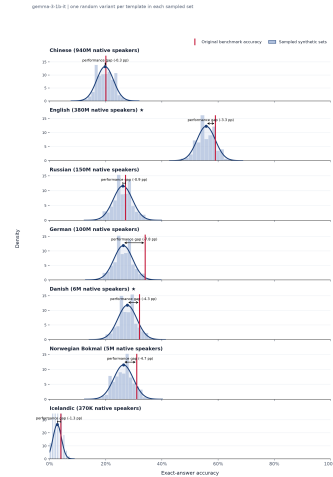
gemma-3-4b-it.png

Original accuracy and synthetic-set distributions



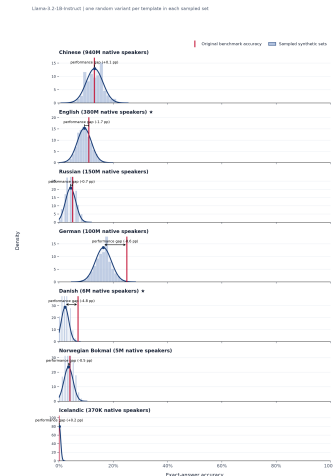
olmo-2-0425-1b-instruct.png

Original accuracy and synthetic-set distributions



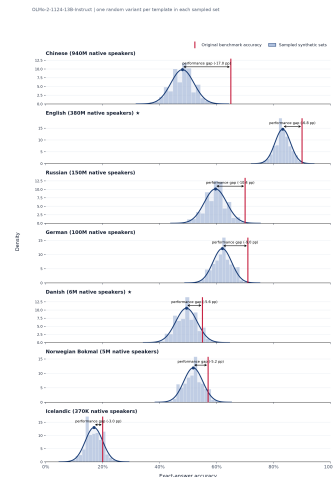
gemma-3-1b-it.png

Original accuracy and synthetic-set distributions



llama-3.2-1b-instruct.png

Original accuracy and synthetic-set distributions



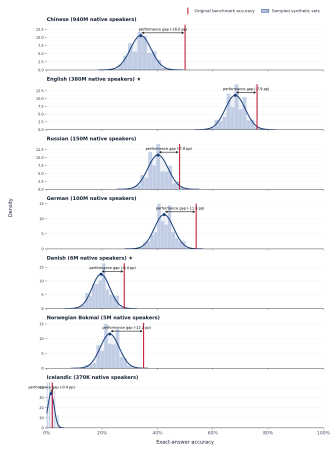
olmo-2-1124-13b-instruct.png

Figure 10: Ridgeline plots, part 1.

918
919
920
921
922
923
924
925
926
927
928
929
930
931
932
933
934
935
936
937
938
939
940
941
942
943
944
945
946
947
948
949
950
951
952
953
954
955
956
957
958
959
960
961
962
963
964
965
966
967
968
969
970
971

Original accuracy and synthetic-set distributions

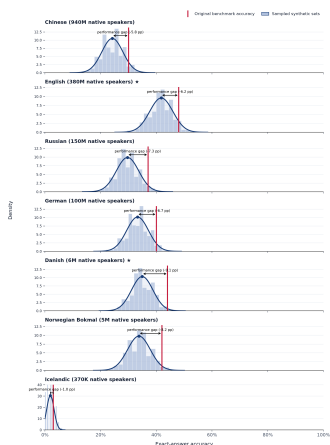
ollm-2-1124-7b-instruct.png



ollm-2-1124-7b-instruct.png

Original accuracy and synthetic-set distributions

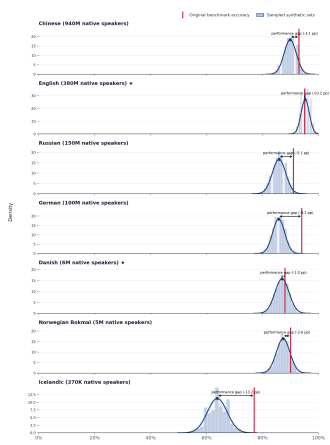
ollm-2-1124-7b-instruct.png



euollm-9b-instruct.png

Original accuracy and synthetic-set distributions

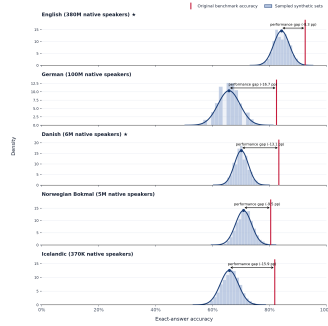
euollm-9b-instruct.png



qwen2.5-14b-instruct.png

Original accuracy and synthetic-set distributions

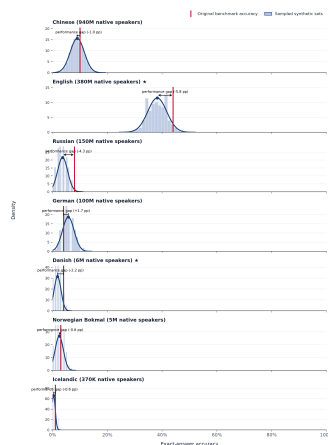
gpt-5.4-nano.png



gpt-5.4-nano.png

Original accuracy and synthetic-set distributions

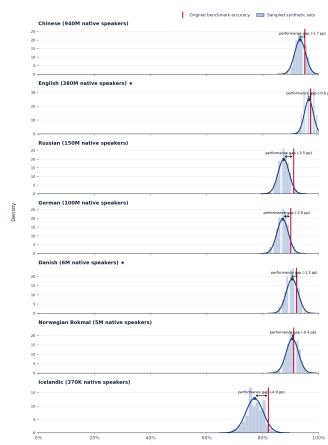
gpt-5.4-nano.png



qwen2.5-0.5b-instruct.png

Original accuracy and synthetic-set distributions

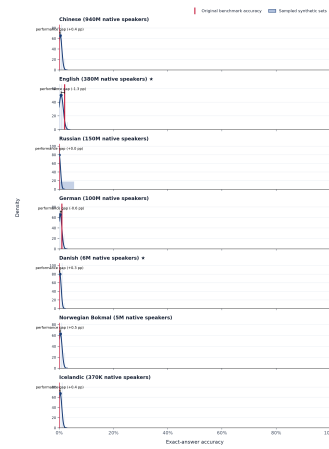
qwen2.5-0.5b-instruct.png



qwen2.5-32b-instruct.png

Original accuracy and synthetic-set distributions

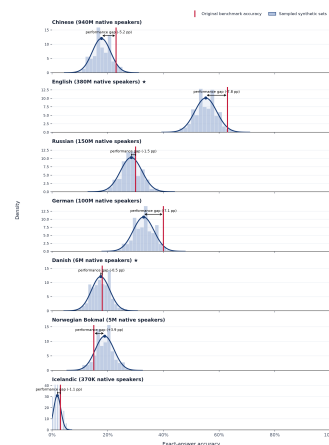
euollm-1.7b-instruct.png



euollm-1.7b-instruct.png

Original accuracy and synthetic-set distributions

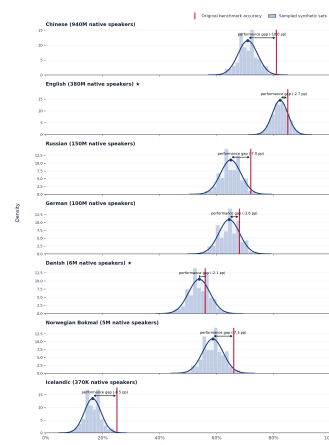
euollm-1.7b-instruct.png



qwen2.5-1.5b-instruct.png

Original accuracy and synthetic-set distributions

qwen2.5-1.5b-instruct.png



qwen2.5-3b-instruct.png

Figure 11: Ridgeline plots, part 2.

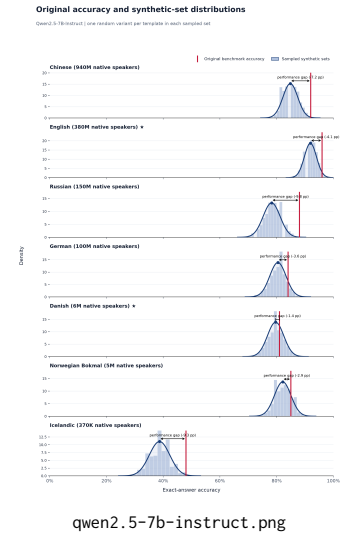
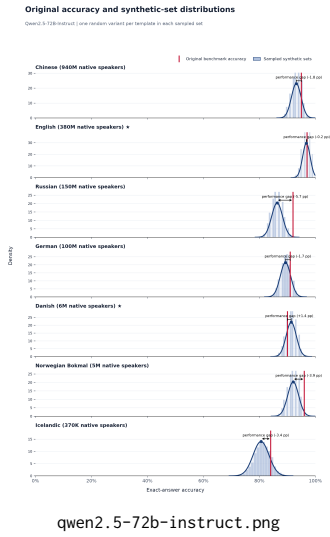


Figure 12: Ridgeline plots, part 3.

This is an appendix.